the universe expands forever, but only barely so. In that case the shape is flat, preserving all the geometric rules we learned in high school about parallel lines. If omega exceeds one, parallel lines converge, and the universe curves back on itself, ultimately recollapsing into the fireball whence it came.

At no time since Hubble discovered the expanding universe has any team of observers ever reliably measured omega to be anywhere close to one. Adding up all the mass and energy their telescopes could see, and even extrapolating beyond these limits, dark matter included, the biggest values from the best observations topped out at about $\Omega = 0.3$. As far as observers were concerned, the universe was "open" for business, riding a one-way saddle into the future.

Meanwhile, beginning in 1979, the American physicist Alan H. Guth of the Massachusetts Institute of Technology, and others, advanced an adjustment to the big bang theory that cleared up some nagging problems with getting a universe to be as smoothly filled with matter and energy as ours is known to be. A fundamental by-product of this update to the big bang was that it drives omega toward one. Not toward a half. Not toward two. Not toward a million. Toward one.

Hardly a theorist in the world had a problem with that requirement, because it helped get the big bang to account for the global properties of the known universe. There was, however, another little problem: the update predicted three times as much mass-energy as observers could find. Undeterred, the theorists said the observers just weren't looking hard enough.

At the end of the tallies, visible matter alone could account for no more than 5 percent of the critical density. How about the mysterious dark matter? They added that, too. Nobody knew what it was, and we still don't know what it is, but surely it contributed to the totals. From there we get five or six times as much dark matter as visible matter. But that's still way too little. Observers were at a loss, and the theorists answered, "Keep looking."

Both camps were sure the other was wrong—until the discovery of dark energy. That single component, when added to the ordinary matter and the ordinary energy and dark matter, raised the mass-energy density of the universe to the critical level. Simultaneously satisfying both the observers and the theorists.

For the first time, the theorists and observers kissed and made up. Both, in their own way, were correct. Omega does equal one, just as the theorists demanded of the universe, even though you can't get there by